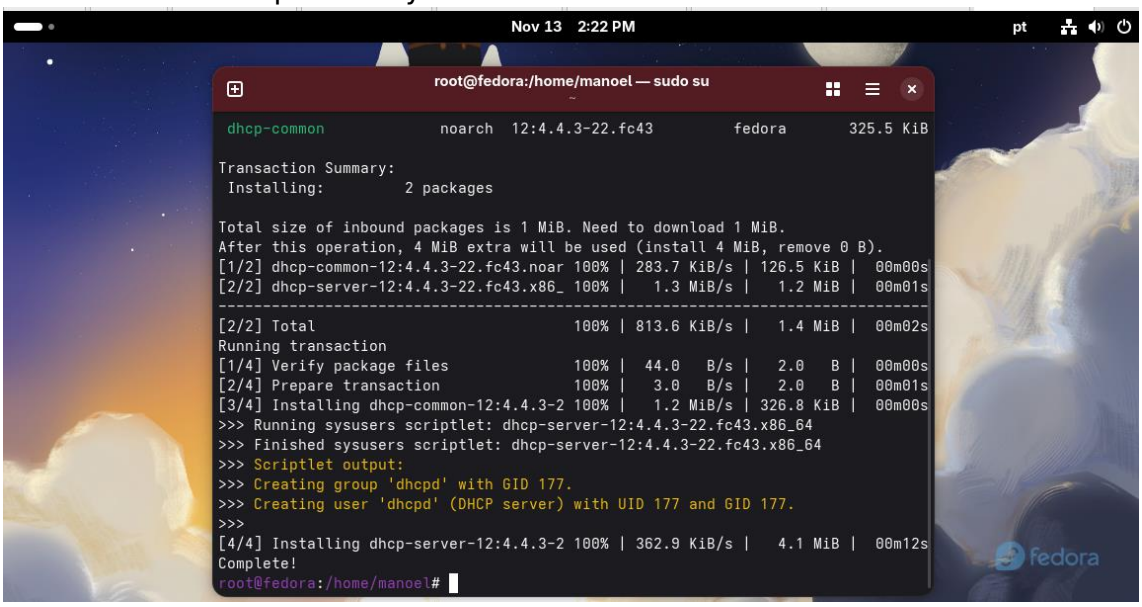
## 1. Instalar e configurar o servidor DHCP

### Instalação:

- sudo dnf install dhcp-server -y

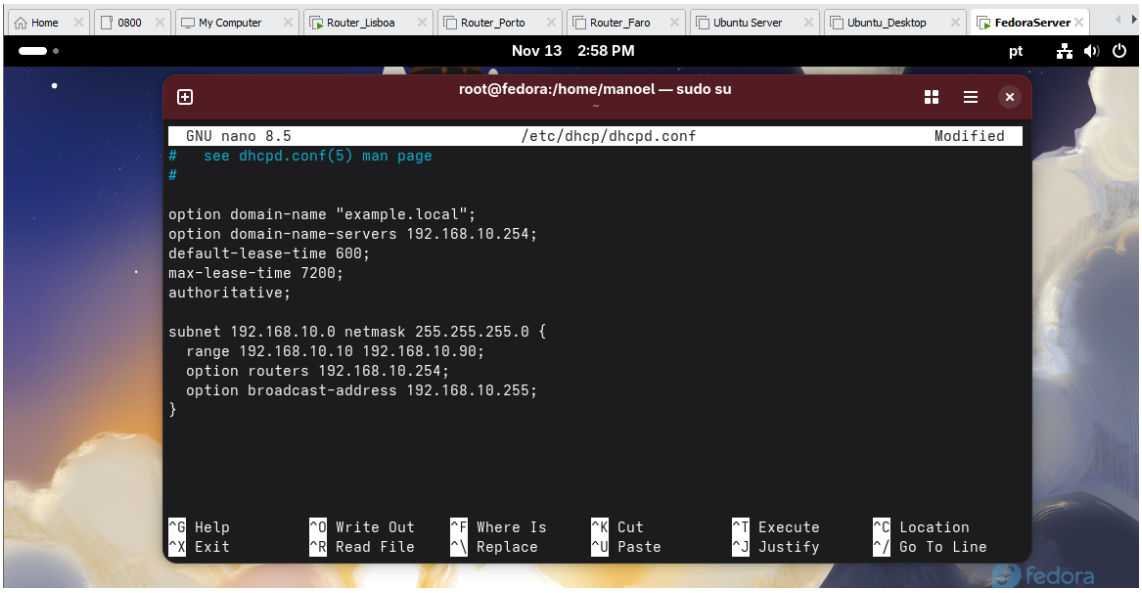



### Configuração:

### Edite o arquivo de configuração:

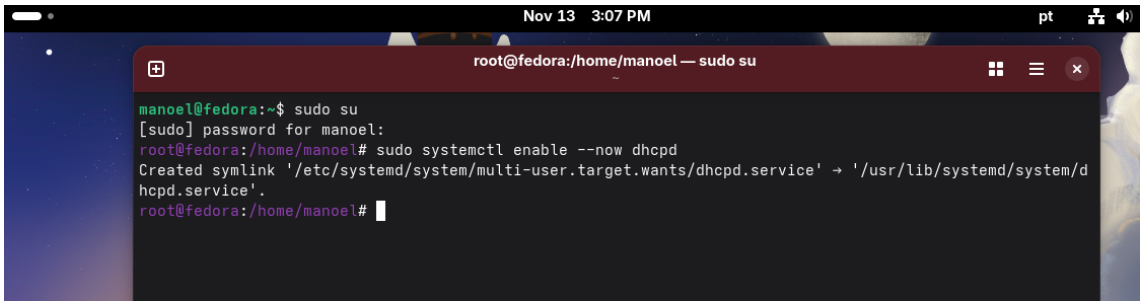
- sudo nano /etc/dhcp/dhcpd.conf

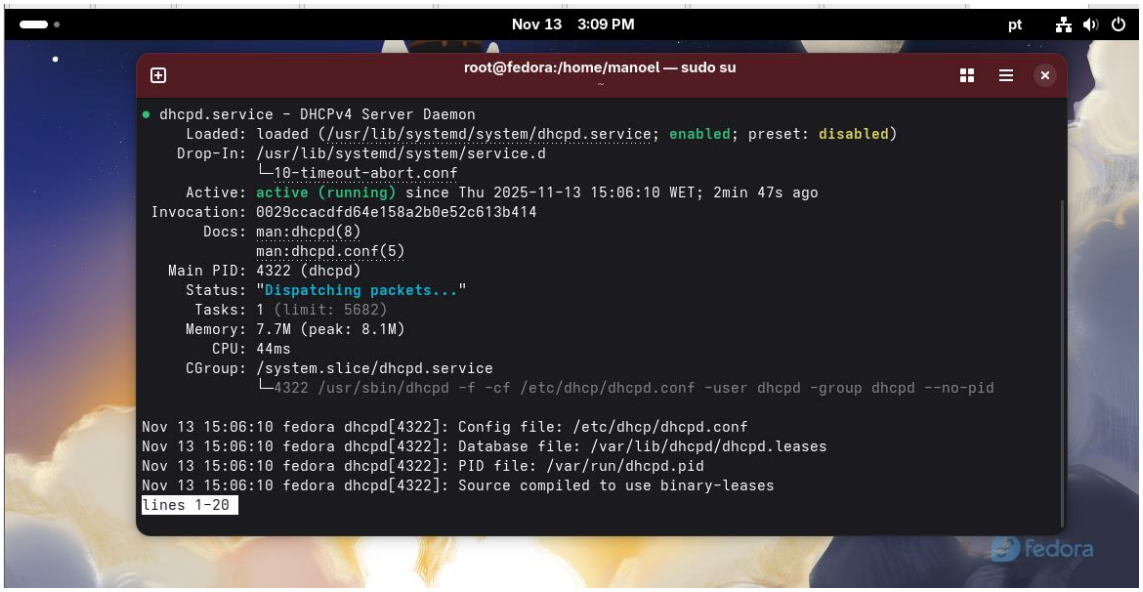
### Exemplo básico:

- 


### Ativação:

- sudo systemctl enable --now dhcpd

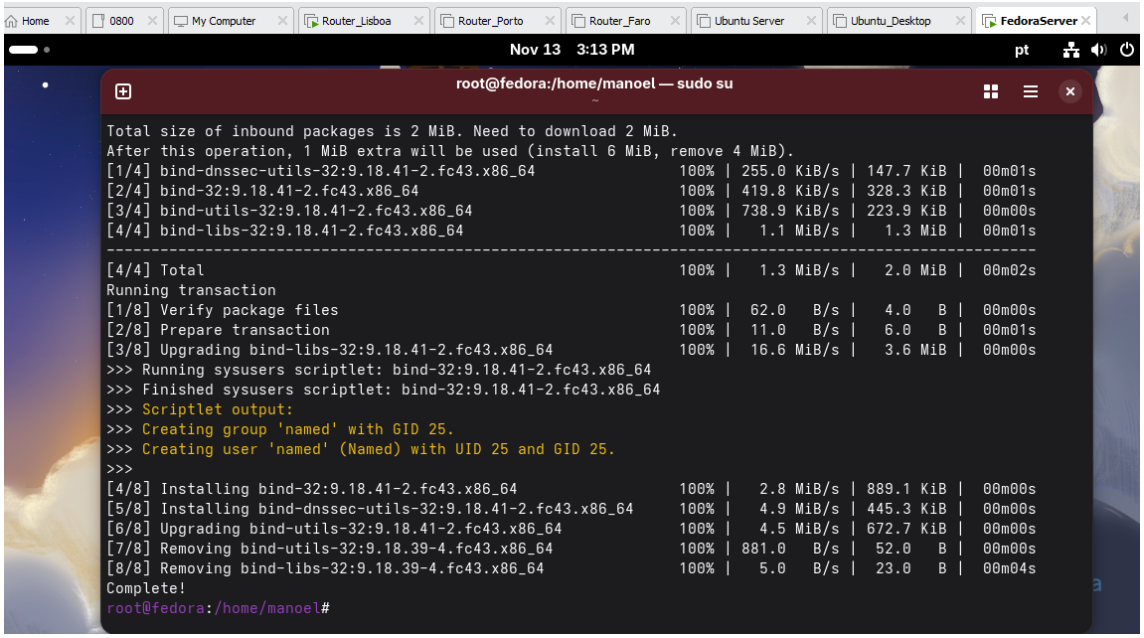



## 2. Instalar e configurar o servidor DNS (Bind)

**Instalação:**

- sudo dnf install bind bind-utils -y



**Configuração:**
**Edite  e crie zonas em .**
**Exemplo de zona direta:**

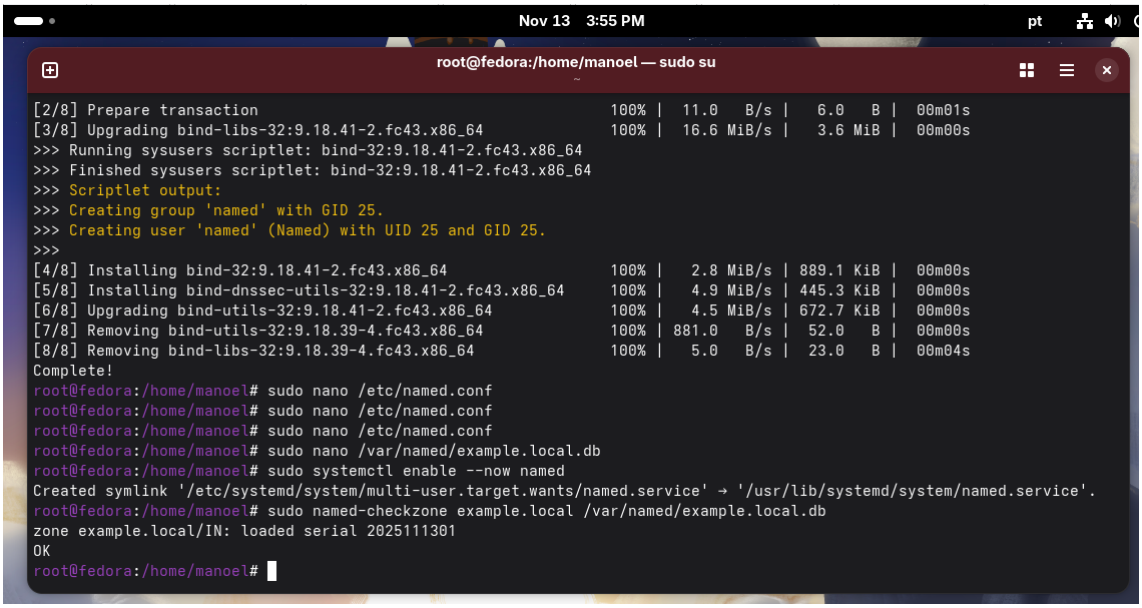
- zone "example.local" IN {
- type master;
- file "example.local.db";
- };

**Exemplo de arquivo de zona ():**

- $TTL 86400
- @ IN SOA ns.example.local. admin.example.local. (
- 2025111301 ; Serial
- 3600 ; Refresh
- 1800 ; Retry
- 604800 ; Expire
- 86400 ) ; Minimum
-
- @ IN NS ns.example.local.
- ns IN A 192.168.1.10
- www IN A 192.168.1.20

**Ativação:**

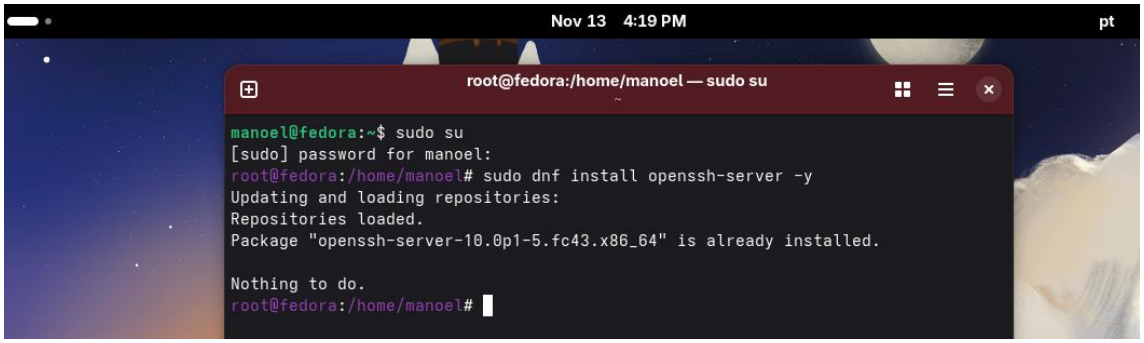
- sudo systemctl enable --now named



## 3. Instalar e configurar o servidor SSH

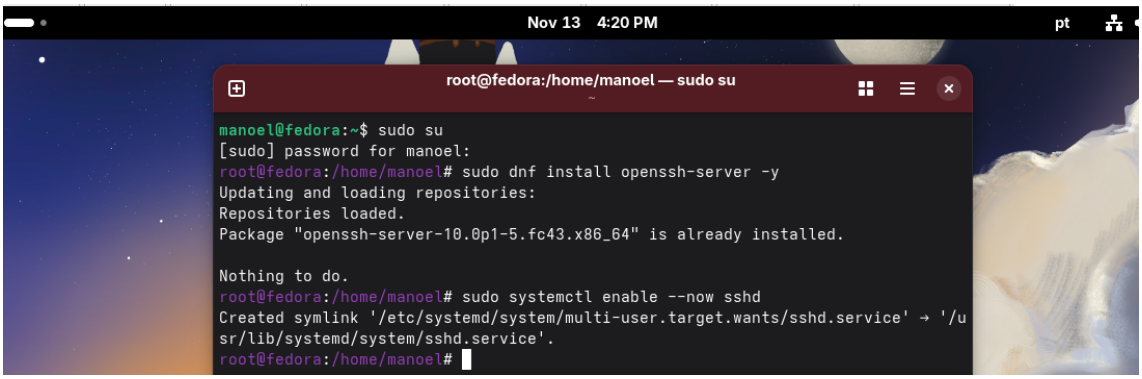
### Instalação:

- sudo dnf install openssh-server -y



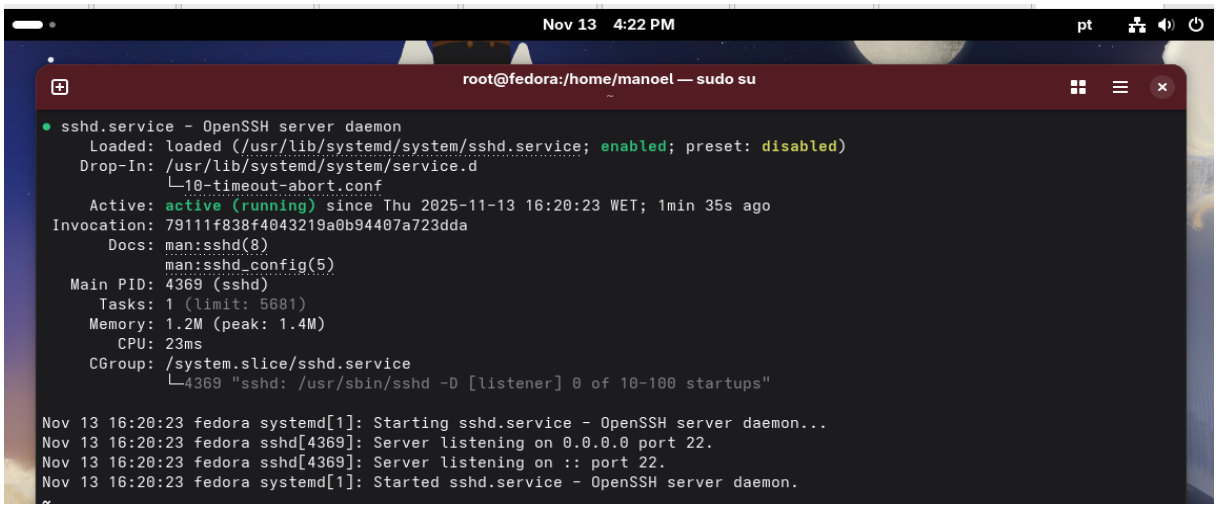

### Ativação:

- sudo systemctl enable --now sshd




### Verificação:

- systemctl status sshd



## 4. Instalar e configurar o servidor Apache (HTTPD)

### Instalação:

- sudo dnf install httpd -y

```
root@fedora:/home/manoel# sudo dnf install httpd -y
Updating and loading repositories:
Repositories loaded.
Package "httpd-2.4.64-3.fc43.x86_64" is already installed.

Nothing to do.
root@fedora:/home/manoel#
```

### Ativação:

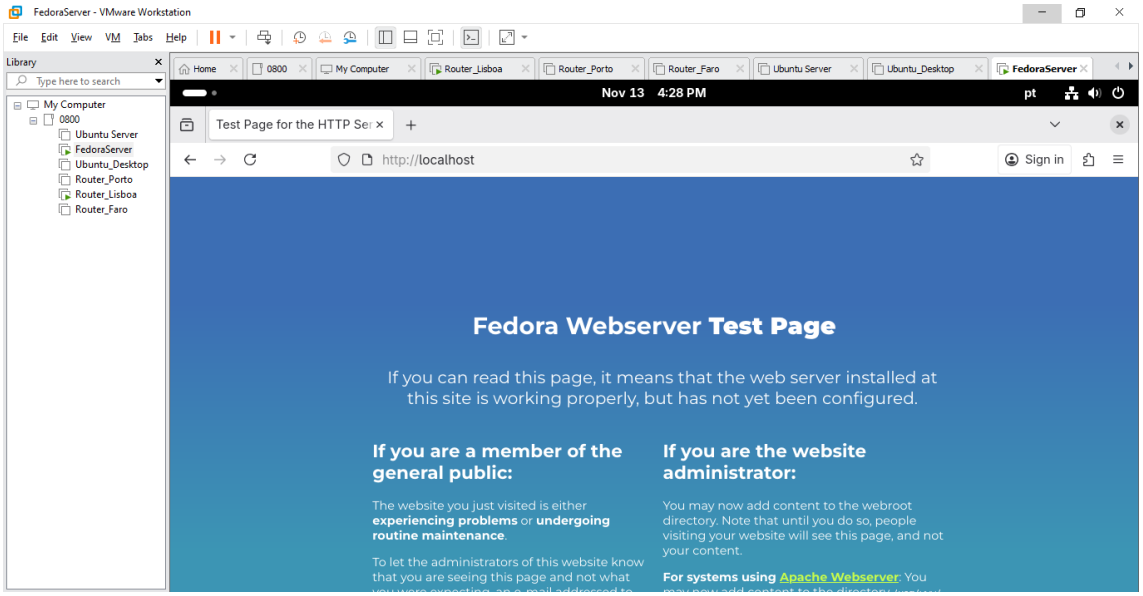
- sudo systemctl enable --now httpd

```
Nothing to do.
root@fedora:/home/manoel# sudo systemctl enable --now httpd
Created symlink '/etc/systemd/system/multi-user.target.wants/httpd.service' → '/usr/lib/systemd/system/httpd.service'.
root@fedora:/home/manoel#
```

### Teste:

### Abra o navegador e acesse  ou .

- Abra o navegador e acesse `http://localhost` ou `http://`




## 5.Firewall (opcional)

### Certifique-se de que os serviços estão liberados no firewall:

- sudo firewall-cmd --permanent --add-service=dhcp

```
manoel@fedora:~$ sudo su
[sudo] password for manoel:
root@fedora:/home/manoel# sudo firewall-cmd --permanent --add-service=dhcp
success
```

- sudo firewall-cmd --permanent --add-service=dns

```
root@fedora:/home/manoel# sudo firewall-cmd --permanent --add-service=dns
success
```

- sudo firewall-cmd --permanent --add-service=ssh

```
root@fedora:/home/manoel# sudo firewall-cmd --permanent --add-service=ssh
Warning: ALREADY_ENABLED: ssh
success
```

- sudo firewall-cmd --permanent --add-service=http

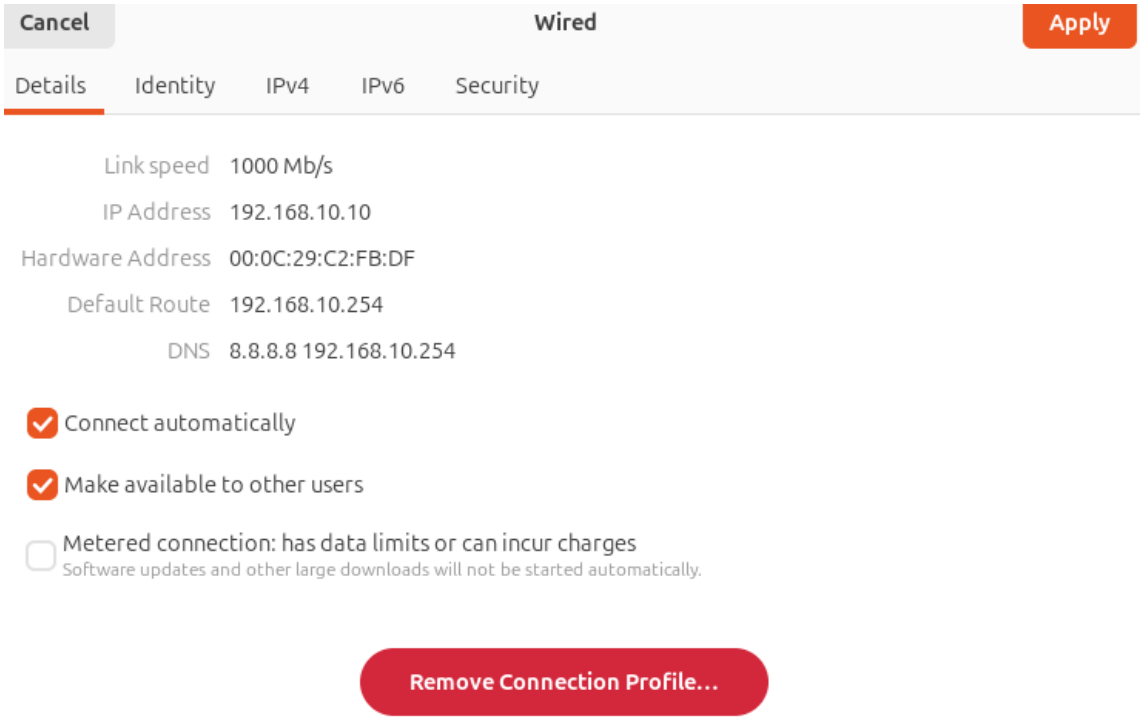
```
root@fedora:/home/manoel# sudo firewall-cmd --permanent --add-service=http
success
```

- sudo firewall-cmd –reload

```
root@fedora:/home/manoel# sudo firewall-cmd --reload
success
root@fedora:/home/manoel#
```

# UBUNTU DESKTOP

**1. Testar o serviço DHCP**
**Verificar se o Ubuntu está a receber IP via DHCP:**
**1.**




**2. Verifique o IP atribuído:**

- ip a

```
Nov 14 14:18
root@manoel-VMware-Virtual-Platform: /home/manoel
manoel@manoel-VMware-Virtual-Platform:~$ sudo su
[sudo] password for manoel:
root@manoel-VMware-Virtual-Platform:/home/manoel# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:c2:fb:df brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.10.10/24 brd 192.168.10.255 scope global dynamic noprefixroute ens33
       valid_lft 435sec preferred_lft 435sec
root@manoel-VMware-Virtual-Platform:/home/manoel#
```

## 🌐 2. Testar o serviço DNS

**Usar  ou :**

```
root@manoel-VMware-Virtual-Platform:/home/manoel# nslookup www.example.local 192.168.10.3
Server:         192.168.10.3
Address:        192.168.10.3#53

Name:   www.example.local
Address: 192.168.10.20

root@manoel-VMware-Virtual-Platform:/home/manoel#
```

## 🔐 3. Testar o serviço SSH

**Conectar ao servidor Fedora via SSH:**

- ssh usuario@192.168.10.10
- **Substitua  pelo nome de utilizador no Fedora.**
- **Se for a primeira vez, aceite a chave e insira a senha.**

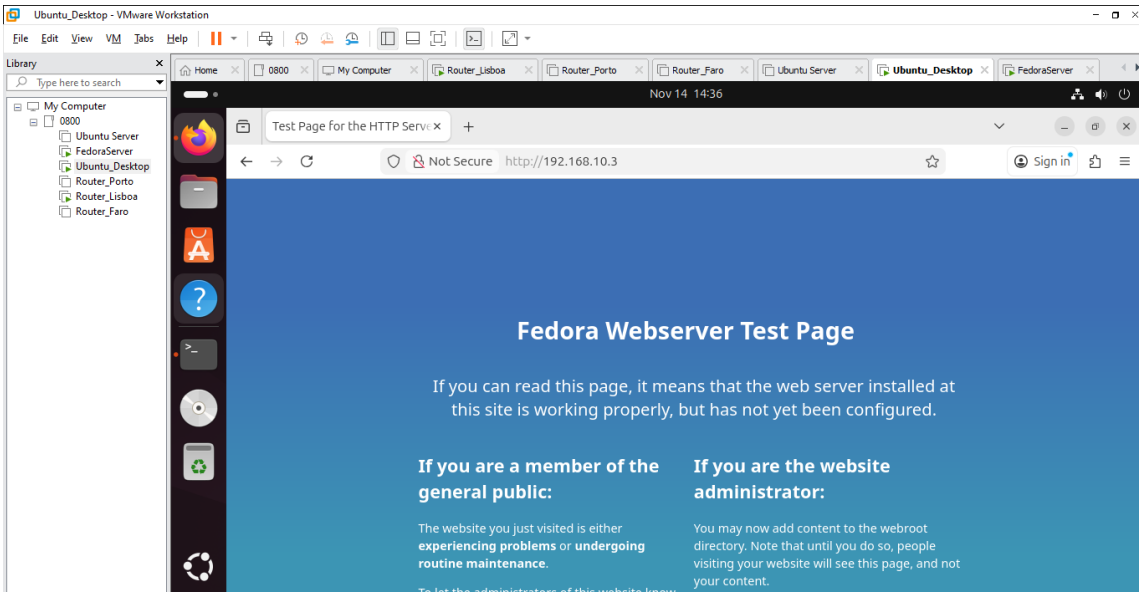
```
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.10.3' (ED25519) to the list of known hosts.
manoel@192.168.10.3's password:
Activate the web console with: systemctl enable --now cockpit.socket

manoel@fedora:~$
```

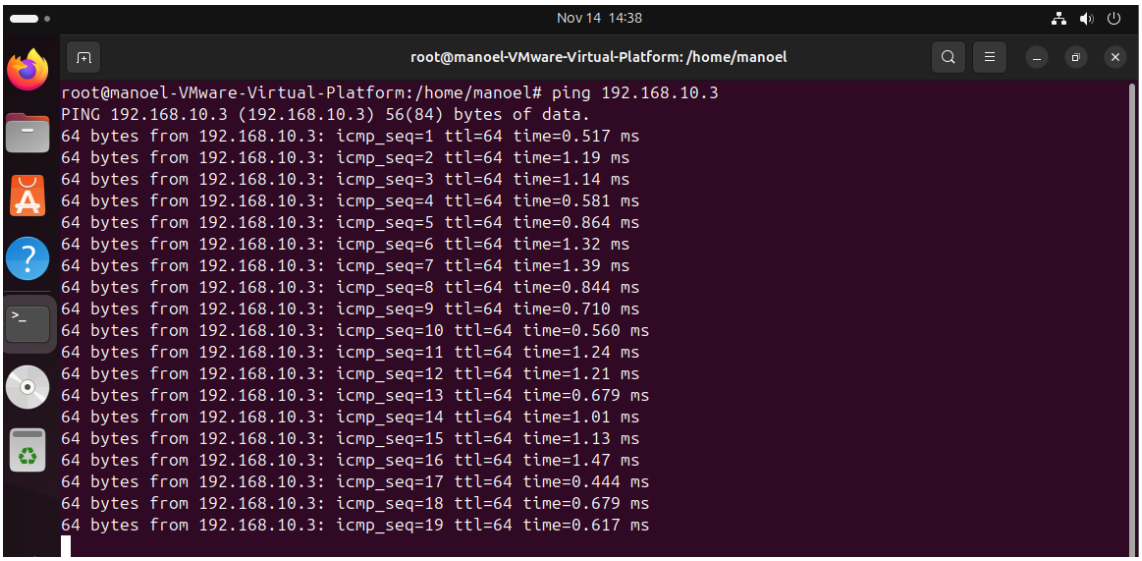
## 🌍 4. Testar o serviço Apache (HTTP)

**Acessar via navegador:**

- **Abra o Firefox ou Chrome no Ubuntu.**




- **Use para testar conectividade:**
  - ping 192.168.10.3




- **Use para verificar portas abertas:**
  - sudo apt install nmap
  - nmap 192.168.10.10

```
root@manoel-VMware-Virtual-Platform:/home/manoel# nmap 192.168.10.3
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-11-14 14:40 WET
Nmap scan report for 192.168.10.3
Host is up (0.00100s latency).
Not shown: 845 closed tcp ports (reset), 145 filtered tcp ports (no-response), 7 filtered tcp ports (admin-p
rohibited)
PORT   STATE SERVICE
22/tcp open  ssh
53/tcp open  domain
80/tcp open  http
MAC Address: 00:0C:29:71:13:FF (VMware)

Nmap done: 1 IP address (1 host up) scanned in 3.01 seconds
root@manoel-VMware-Virtual-Platform:/home/manoel#
```